



AIMS

African Institute for
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RWANDA

Introduction to Probability and Statistics

Lecture 8: Point Estimation

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2 Method of Moments

3 Maximum Likelihood Method

Introduction

- We observe n r.v.'s $X_1, \dots, X_n$ (not necessarily independent).
Suppose that $X_i \sim f(x \mid \theta)$ where θ is an **unknown parameter**.
- The problem of estimating θ is a **problem of point estimation**.

Definition

- Any function of the r.v.'s $X_1, \dots, X_n$ is called a **statistic**.
- An **estimator** of θ is a statistic of $X_1, \dots, X_n$:

$$g_n(X_1, \dots, X_n).$$

Estimator

- Since $X_1, \dots, X_n$ are r.v.'s, so is $g_n(X_1, \dots, X_n)$.
- In general $\theta \in \Theta \subseteq \mathbb{R}^q$, that is $\theta = (\theta_1, \dots, \theta_q)$.

Examples: normal distribution: $\theta = (\mu, \sigma^2)$, Gamma distribution: (α, β) .

- Vocabulary:
 - Estimator: $g_n(X_1, \dots, X_n)$.
 - Estimate: $g_n(x_1, \dots, x_n)$ (value).
- How could we build estimators?
 - ➡ Methods of estimation?

Empirical moments v. Theoretical moments

Let us consider n r.v.'s $X_1, \dots, X_n$.

- Empirical moment of order k :

$$m_k = \frac{1}{n} \sum_{i=1}^n X_i^k \quad (\text{statistic}).$$

- Theoretical moments of order k

$$E(X^k; \theta) = \begin{cases} \sum_{x \in V} x^k P(X = x | \theta) & \text{if } X \text{ is discrete.} \\ \int_V x^k f_X(x | \theta) dx & \text{if } X \text{ is continuous.} \end{cases}$$

Method of Moments Estimators (MME)

The method of moments estimators (MME) of $\theta_1, \dots, \theta_q$ are the solution of the equations:

$$m_k = E(X^k; \theta_1, \dots, \theta_q); \quad k = 1, \dots, q.$$

Thus, if the parameter to be estimated is a q -dimensional vector, the method of moments propose to equate the first q empirical moments to the first q theoretical moments and to consider the solution with respect to $\theta_1, \dots, \theta_q$ as estimator of $\theta_1, \dots, \theta_q$.

Exercise: Compute the MME of the parameter of the following model given that a n -sample $X_1, \dots, X_n$ is observed:

- $\mathcal{B}(N, p)$,
- $\mathcal{N}(\mu, \sigma^2)$,
- $\mathcal{G}(\alpha, \beta)$.

Method of Moments

Distribution	Parameter	MME
Bernoulli	p	$\hat{p} = \bar{X}_n$
Binomiale	(N, p)	$\hat{p} = \bar{X}_n / N$
Poisson	λ	$\hat{\lambda} = \bar{X}_n$
Exponential	θ	$\hat{\theta} = \bar{X}_n$
Gamma	(α, β)	$\hat{\alpha} = \bar{X}_n^2 / s_X^2 \quad \hat{\beta} = \bar{X}_n / s_X^2$
Normal	(μ, σ^2)	$\hat{\mu} = \bar{X}_n \quad \hat{\sigma}^2 = s_X^2$

Maximum Likelihood Method

- We observe n independent r.v.'s $X_1, \dots, X_n$.

Consider $f(x_1, \dots, x_n \mid \theta)$ or $P(X_1 = x_1, \dots, X_n = x_n \mid \theta)$, the distribution of $X_1, \dots, X_n$, where θ is an **unknown parameter**.

- $f(x_1, \dots, x_n \mid \theta)$ or $P(X_1 = x_1, \dots, X_n = x_n \mid \theta)$ can be seen as the chance of occurrence of $x_1, \dots, x_n$.

➡ This is a function depending on θ which is an unknown parameter.

- A strategy to provide an estimator is to find the value of θ that maximize this quantity.

➡ What has been observed is the more likely.

Maximum Likelihood Estimator

Definition

Let $(X_1, \dots, X_n)$ be n independent r.v.'s with distribution $f(x_1, \dots, x_n | \theta)$ (or $P(X_1 = x_1, \dots, X_n = x_n | \theta)$), where θ is an unknown parameter.

- The **likelihood function** is:

$$L(\theta) = \begin{cases} f(x_1, \dots, x_n | \theta) & (\text{continuous}). \\ P(X_1 = x_1, \dots, X_n = x_n | \theta) & (\text{discrete}). \end{cases}$$

- Any estimator $g_n(X_1, \dots, X_n)$ such that

$$L(g_n(X_1, \dots, X_n)) = \max \{L(\theta); \theta \in \Theta\}$$

is called a **maximum likelihood estimator (MLE)**.

Maximum Likelihood Method

In practise, the procedure to obtain the MLE is the following:

- 1 Compute the likelihood:

$$L(\theta) = f(x_1, \dots, x_n | \theta) \left[= \prod_{i=1}^n f(x_i | \theta) \text{ when independence} \right]$$

$$\text{or } L(\theta) = P(X_1 = x_1, \dots, X_n = x_n | \theta), \theta \in \Theta \subseteq \mathbb{R}^q.$$

- 2 Calculate the log-likelihood:

$$\log L(\theta).$$

- 3 Calculate the log-likelihood equations:

$$\frac{\partial}{\partial \theta_i} \log L(\theta) = 0, \quad i = 1, \dots, q.$$

- 4 Solve these equations :

$$\hat{\theta}_n = \underset{\theta \in \Theta}{\operatorname{argmax}} \log L(\theta).$$

Maximum Likelihood Method

Example : Binomial

Let consider n independent r.v.'s with a binomial distribution with parameter (N, p) .

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Maximum Likelihood Method

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$$L(p) = \prod_{i=1}^n \binom{N}{x_i} p^{\sum_{i=1}^n x_i} (1-p)^{nN - \sum_{i=1}^n x_i}.$$

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Maximum Likelihood Method

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$$\log L(p) = \sum_{i=1}^n \log \binom{N}{x_i} + n\bar{X}_n \log p + n(N - \bar{X}_n) \log(1-p).$$

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$$\frac{\bar{X}_n}{p} - \frac{N - \bar{X}_n}{1-p} = 0.$$

④ Solution:

Maximum Likelihood Method

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$$\hat{p} = \bar{X}_n / N.$$

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Remark: The MLE is the same as the MME.

Maximum Likelihood Method

Example : Exponential

Let consider n independent r.v.'s with an exponential distribution with parameter θ .

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1 Likelihood

$$L(\theta) = \left(\frac{1}{\theta}\right)^n \exp \left\{ -\frac{1}{\theta} \sum_{i=1}^n X_i \right\}.$$

2 Log-Likelihood

$$\log L(\theta) = -n \log \theta - \frac{1}{\theta} \sum_{i=1}^n X_i.$$

3 Likelihood equation:

$$-n + \frac{1}{\theta} \sum_{i=1}^n X_i = 0.$$

4 Solution:

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n.$$

Remark: The MLE is the same as the MME.

Maximum Likelihood Method

Exemple : Normal distribution

Maximum Likelihood Method

Exemple : Normal distribution

Let consider n independent r.v.'s with a normal distribution with parameters (μ, σ^2) .

① Likelihood: $L(\mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right\}.$

② Log-Likelihood: $\log L(\mu, \sigma^2) = -n \log(\sqrt{2\pi}) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$

③ Likelihood equations:

$$\begin{cases} \frac{\partial}{\partial \mu} \log L(\mu, \sigma^2) = 0 \\ \frac{\partial}{\partial \sigma^2} \log L(\mu, \sigma^2) = 0 \end{cases} \Leftrightarrow \begin{cases} \sum_{i=1}^n x_i - n\mu = 0 \\ -n + \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 = 0 \end{cases}$$

④ Solution: $\hat{\mu}_n = \bar{x}_n$ and $\hat{\sigma}^2 = s_x^2$.